NOMVULA CHALOTTE MOKWENA
Exercise 2

The students Table

1. Syntax : SELECT DISTINCT department,
FROM students ;

Table students

Department
IT
HR
Finance

2. Syntax : SELECT department, AVG (age) AS avg_age
FROM students
GROUP BY department;

Table students	
Department	Avg_age
IT	23,5
HR	22.0
Finance	23.0

3. Syntax : SELECT department, COUNT (*) AS student_count
FROM students
GROUP BY department
HAVING COUNT (*) > 1 ;

~~Table~~

Table students

Department	Student_count
IT	2
HR	2

4. Syntax : SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;

Table students

Student_id	Name	Age	Department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. Syntax : SELECT student_id, name, age, department
FROM students
WHERE department IN ('IT', 'HR') AND age > 21;

Table students

Student_id	name	age	Department
1	Alice	26	IT
2	Bob	22	HR
5	Eve	22	HR

6. Syntax : SELECT department, Sum(credits) AS total credits
FROM courses
GROUP BY department
HAVING Sum (credits) > 5;

Table ~~students~~ Courses.

Department	total_credits
IT	11

7. Syntax: SELECT course_id, course_name, department, credits
FROM courses
WHERE credits != 4;

Table courses

course_id	Course_name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. Syntax: SELECT course_id, course_name, credits
FROM courses
ORDER By credits DESC
LIMIT 3;

Table courses

Course_id	course_name	credits
102	Python	4
103	Data Scienc	4
101	SQL basics	3

9. Syntax: SELECT MAX(grade) AS max_grade; MIN(grade) AS min_grade,
AVG(grade) AS avg_grade
FROM enrollments;

Table enrollments

max_grade	min_grade	avg_grade
90	78	84.6

10. Syntax: SELECT course_id, COUNT(*) AS enrollment_count
FROM enrollments
GROUP BY course_id;

Table enrollments

Course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

11. Syntax: SELECT department, SUM(salary) AS total_salary,
SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;

Table salaries

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12. Syntax: SELECT department, AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) >55000;

Table salaries

department	avg-salary
IT	61000
Finance	70000

13. Syntax : SELECT employee_id, name, salary, bonus, salary + bonus AS total_compensation
FROM salaries
WHERE salary + bonus >60000;

Table salaries

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

14. Syntax : SELECT department, Sum(budget) AS total-budget,
AVG(budget) AS avg budget
FROM projects
GROUP BY department
HAVING AVG (budget) >70000;

Table projects

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80000

15. Syntax : SELECT project_id, project_name, department, budget
FROM projects
WHERE budget BETWEEN 50000 AND 120000 AND
department != 'Marketing';

Table projects

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll system	Finance	80 000
5	HR portal	HR	50 000